

Experience

Research Intern - University of Hull - Hull (AUGUST 2022 - NOVEMBER 2022)

KEY RESPONSIBILITIES:

- Designed and developed custom Python software for our in-house-built hybrid 3D printer for printing embedded electronics.
- Created and maintained development documentation to support ongoing software development post-internship, including UI designs for future features created in Justinmind.
- Planned new features to be developed using agile development methodology, including the use of sprints and Kanban boards to track progress.

KEY ACHIEVEMENTS:

- Expanded the initial project's scope by developing custom Python software for gcode manipulation.
- Drastically reduced preparation time for custom gcode files for conductive ink and plastic printing.
- Reduced the error rate and streamlined the process of embedding custom electronic tracks into printed parts.
- Helped increase the possible complexity of the devices that can be manufactured.

Junior Front-End Software Engineer - Global View Systems - Hull (OCTOBER 2021 - MARCH 2022)

KEY RESPONSIBILITIES:

- Designed, developed, and maintained responsive front-end features for key software using Blazor WebAssembly and contributed to mobile development using Flutter.
- Developed and optimised back-end solutions, including data migrations and schema design, using Entity Framework and .NET Core to enhance data management efficiency. Helped the in-house testing team with the deployment of software on Microsoft Azure.
- Used .NET to develop new features for our new auditing software targeted at the healthcare industry.
- Used .NET MVC to develop and maintain features for our more mature porter software for NHS hospital use, including working with radio integration to the software (Motorola, Hytera) and maintained mobile applications on Android and iOS.

KEY ACHIEVEMENTS:

- Implemented a full-stack employee sign-in system for fire safety in the office, including a dynamic backend to accommodate future hires and a front end using the Blazor UI framework MudBlazor.
- Implemented new time-tracking features, allowing management to streamline audit allocation and keep track of upcoming audits, including yearly audits such as efficacy audits.

Current Education

University of Hull - MSc Research Masters in Physics (Nanoelectronics) (JANUARY 2023 - JULY 2025)

- Expanded the scope of the initial software tool created during my internship to include the creation of custom gcode.
- Developed a 2D line-by-line simulated gcode visualiser with the help of DearPyGui.
- Implemented a feature set that enables seamless operation of the modified 3D printing process without user input or overhead.
- Conducted hands-on testing of the software with the printer, including both conductive ink and plastic printing.
- Developed a feature enabling the integration of any gcode file with various infill settings for compatibility with the hybrid 3D printer.
- Maintained and tested the custom 3D printer, including designing custom inserts and add-ons in CAD and manufacturing them for better print quality.

Projects

EEG-Controlled Prosthetic

Used Python to develop software that can control a 3D-printed prosthetic by reading real-time brain data. The software mimics an EEG brain-reading device, outputting data similarly to brains. The other part of the software reads in the data with the help of sockets; the software then processes it into a graph using Matplotlib.

Efficient STL Searching Tool

I am currently working on a web-based STL searching tool that will allow users to search through archive file formats such as .zip or .7z. The tool consists of two components: a searching component and a viewing component. The viewer is a full 3D viewing system, such as CAD or 3D printing slicer software, allowing the user to view each STL file in 3D space. The tool is written in React.js and Three.js and is currently in the optimisation phase of development.

Skills

Programming Languages

C - C++ - C# - Python - JavaScript - SQL - HTML - CSS - Zig - TypeScript

Frameworks

.NET Core - Blazor - React - Node.js - Svelte - Vue - HTMX

Tools

Three.js - OpenGL - SFML - Git - Entity Framework - Jira - Docker - FastAPI - jQuery - Bootstrap - Arduino - MySQL - PostgreSQL - Supabase - DearPyGui - Fusion 360 - Blender - Adobe CC - Ultimaker Cura - NumPy - Matplotlib - CAD - Figma - Justinmind - Flutter - Microsoft Azure

Experimental Skills

3D Printing - Chemical Cleaning - General Electronics - Microscopic Photography - Profilometer

Languages

English - Arabic - Swedish

Education

University of Hull - First-Class Honours - BSc Computer Science (SEPTEMBER 2018 - MAY 2021)

- Advanced Programming (93%): Focused on advanced software features within object-orientated languages. Utilised C++ to understand memory access, work with pointers, and perform code analysis and testing with tools like Parasoft.
- Artificial Intelligence (72%): Developed a genetic algorithm for an existing neural network in C#.
- System Analysis, Design, and Process (71%): Group project to design and develop a piece of software and go through the entire software development process using the Agile methodology.
- Agile Software Development (70%): Developed a forum website in a group using Scrum and the Agile methodology.
- Electronics and Interfacing (69%): Developed an assortment of various small programs for an Arduino microcontroller.
- Object-Orientated Programming (60%): Created the game "Uno" in C# using object-orientated design and methodologies.

Hull College - Distinction* Distinction Distinction - Level 3 Extended Certificate in IT (SEPTEMBER 2016 - JUNE 2018)

- Database Development - Worked in a team to design and develop a database for a movie-booking website in Microsoft Access.
- Computer Games Design - Designed and developed a 2D platformer game for the Unity Game Engine, including creating design documents and a game guide.
- Organisational Systems Security - Designed security measures for a given computer system for a mid-sized business.

Teaching

Laboratory Demonstrator - University of Hull Department of Physics - Hull (SEPTEMBER 2023 - DECEMBER 2025)

KEY RESPONSIBILITIES:

- Helped students with their experimental methodology, data analysis, and Arduino scripts.
- Guided students in developing their analytical thinking skills, teamwork skills, and lab skills.
- Ensured students were compliant with lab safety by having them set up and follow lab risk assessments.
- Marked and provided feedback for students and their assignments.

Volunteering

University of Hull (JULY 2025 - Current)

- Continue with my masters research. Working on more complex devices with the help of the software and custom 3D printer
- Implemented more advanced homing and nozzle wiping features for the machine.
- Started exploring options for printing electronic tracks in 3 dimensions.

Smash Crab Studios (SEPTEMBER 2015 - DECEMBER 2015)

- Gained an introduction to the Unity Game Engine and basic game development principles.
- Worked in a team to design abilities for a mobile game.
- Learnt basic Git commands and source control principles.
- Practised leadership skills by leading a small team to create our unique take on the game Snake with more realistic movements in the Unity Game Engine.